

Requirements Specification - Rising Waters

1. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent (minimum); Intel i5 or above (recommended for model training).
- RAM: 4 GB (minimum); 8 GB (recommended for running local notebooks and training pipelines).
- Storage: 2 GB free disk space (to house datasets, virtual environments, and source code).
- Network: Active internet connection for Vercel/GitHub deployment and initial library installations.

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Development Environment: VS Code or PyCharm.
- Python Runtime: Python 3.8 to 3.12.
- Web Framework: Flask (WSGI web micro-framework).
- Web Libraries: Bootstrap 5, Bootstrap Icons, FontAwesome.
- ML Libraries (Local Development): Scikit-learn, XGBoost, Pandas, Numpy, Openpyxl.

2. Data Flow Diagrams (DFD)

Level 0 DFD: Context Diagram

mermaid

graph TD

User([Meteorologist / Coordinator]) -->|Input Weather Parameters| System[[Rising Waters System]]

System -->|Flood Warning / Safe Status| User

System -->|Confidence Score %| User

Level 1 DFD: System Architecture

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graph TD

User([User]) -->|POST Web Form| App[Flask Server app.py]

App -->|Input Sanitization| Val[Validation Engine]

Val -->|If Errors| Flash[Flashed Error Message]

Flash -->|Redirect| User

Val -->|If Valid| Scale[Manual Scaling scaling_loop]

Scale -->|Z-Score Features| Model{Decision Tree Classifier}

Model -->|Binary Class| Predict[Prediction Result]

Model -->|Sigmoid Confidence| Prob[Confidence Score %]

Predict & Prob -->|Render Outcome| Result[chance / no_chance Card]

Result -->|HTML Display| User